

SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, VIII SEM

Date : 4th APR 2016, Monday

Time : 02:30pm To 05:30pm

Session : Evening

Sub: Artificial Intelligence

Course No : PS08CIIT03

Total marks : 70

Q – 1 Select an appropriate option

[08]

- i) The test in which the computer mimics a human and tries to fool other humans is called _____.
 a) Turing Test b) Chinese room Test c) AI Test d) None of These
- ii) _____ includes the published definitions facts and theories.
 a) Public Knowledge b) Private Knowledge
 c) Expert Knowledge d) None of These
- iii) Factual knowledge uses _____ symbol for OR.
 a) ^ b) □ c) ~ d) ==>
- iv) _____ is the type of knowledge acquisition in which the domain expert is the only source of knowledge.
 a) Knowledge Engineer b) Knowledge Elicitation
 c) Domain Expert d) None of above
- v) _____ maps elements of a fuzzy set to real numbered values in the interval 0 to 1.
 a) Fuzzy Logic b) Bivalued Logic c) Membership Function d) None
- vi) In fuzzy connectives implications are obtained by _____.
 a) Union b) Intersection c) Negation & Intersection d) Negation
- vii) _____ is a collection of such genes representing possible solutions in domain space
 a) Genetic Cycle b) Genetic algorithm c) Mutation d) Genotype
- viii) To evaluate strength of an individual _____ is used in GA.
 a) Genotype b) Optimization c) Encoding Strategy d) Fitness Function

Q – 2 Answer the following Questions (Any SEVEN)

[14]

- i) Explain source of knowledge
- ii) Draw a data pyramid for managerial perspectives.
- iii) What is knowledge base? and list components of it.
- iv) List Difficulties in KBS Development explain any one.
- v) Draw a diagram of development methodology for a Knowledge Based System.
- vi) Draw a figure of Fuzzy controller
- vii) Write probabilistics Neural Networks.
- viii) Write Genetic Algorithm steps
- ix) Draw a figure of Genetic Cycle.

Q – 3 a) Define KBS and Explain categories of KBS

[06]

b) Explain Advantages of Knowledge Based System.

[06]

OR

Q – 3 b) Write difficulties adjoin with the KBS.

[06]

- Q – 4 (a) Explain existing techniques for knowledge acquisition. [06]
 (b) Explain representing procedural knowledge. [06]
- OR**
- Q – 4 (b) Explain Knowledge based system tools [06]
- Q – 5 (a) Explain Fuzzy Propositions and Fuzzy Inference. [06]
 (b) Explain Neural network architectures. [06]
- OR**
- Q – 5 (b) Write Applications of Neural Networks in detail. [06]
- Q – 6 (a) Write introduction of Genetic Algorithms with basic terminology [06]
 (b) Explain basic operators of a Genetic Algorithm. [06]
- OR**
- Q – 6 (b) Minimize a simple function $f(x)=x*x$ on the integer interval [0,30] using genetic algorithm. Take the 01101, 11000, 01000, 10011 strings as individuals in the first population [06]

$$x' = x = x$$